

LESSON 112 (0.5 GROUND and 1.3 DUAL)**Lesson Objective**

During this lesson, the student will become more familiar with the skills and risk management associated with stalls. Additionally, the student will be introduced to various configurations and phases of flight in which stalls can occur as well as turns around a point.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Ask the students if they have any questions on the procedures from the maneuver's manual.
 - Power-off stall.
 - Power-on stall.
 - Turns around a point.
 - Have the student draw out turns around a point to the best of their understanding.
 - Quiz the student on the reading from the Airplane Flying Handbook as needed.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
 - Stall and Spin Awareness Training. ([AC 61-67](#))
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Chapter 07 - Ground Reference Maneuvers.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 05 - Aerodynamics of Flight.

Lesson Content (Ground)**Stalls**

- Can occur at any pitch, any attitude or airspeed.
- Wing does not stop producing lift – it cannot generate adequate lift to sustain level flight.
- Wing designed to stall at wing root.
- Three most common situations where critical AoA can be exceeded:
 - Low Speed.
 - High speed.
 - Turning.
- Explain how stall speed is directly affected by load factor and bank angle.

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- Load factor = $1 / \cos$ (angle of bank).
- Stall speed banked = stall speed level / square root of load factor.
- It is important for the student to understand stall characteristics as well as indications of a stall in order to provide the proper recovery procedures.
 - Indications:
 - Stall warning.
 - Buffeting.
 - Slow airspeed.
 - Reduced control effectiveness.
- Recovery procedures must be prompt and positive in order to achieve minimal loss of altitude.
- Quiz the student on the procedures for a power-off stall and a power-on stall, as this should have been reviewed for homework.

Turns Around a Point

- The student should follow along with the Airplane Flying Handbook chapter 6.
- Ground reference maneuver that consists of a 360 degrees (x2) constant radius turn around a ground-based reference point.
- Select a prominent ground-based reference point.
 - Mention obstacle avoidance and landing point in case of emergency.
- Enter on the downwind at 1,000 feet.
 - Use smoke, water, PFD, groundspeed, etc. to determine where the winds are coming from.
 - Mention how close/far away from the point the entry should be.
 - Emphasize space and time to set up!
- Higher groundspeed requires a steeper bank and vice-versa. The bank angle will be adjusted throughout the maneuver.
- Objectives of the maneuver:
 - Adjusting the bank angle accordingly in correlation with the groundspeed.
 - Maintaining a specific distance between the airplane and the reference point.
 - Division of attention.
 - Compensating for wind drift.
- Common errors:
 - Failure to clear the area.
 - Failure to enter the maneuver at a constant altitude and airspeed.
 - Failure to maintain altitude during the maneuver.
 - Failure to assess wind direction.
 - Failure to maintain a constant radius turn.
 - Failure to maintain airplane coordination.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.

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- FIT minimums.
- Aircraft.
- Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal/Crosswind Takeoff

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

Review Previous Exercises as required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Effects of Power and Flap on a Stall

- Describe why there is a difference between V_{s1} and V_{s0} .
- Introduce the concept of the accelerated stall:
 - Show the formula (accelerated stall speed = $V_s * \sqrt{\text{load factor}}$).

Power-Off Stall

- First completion of the maneuver will be a demonstration.
 - Instructor does, student says.
 - Verify correct procedure.
 - Stress smoothness and coordination.
 - Explain how an improper recovery can lead to a secondary stall.
- Second completion of the maneuver will be performed by the student.
 - Student says, student does.
 - Verify correct procedure.

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- Stress smoothness and coordination.
- Guard against a secondary stall.
- Provide guidance for the student as needed.

Cross-Controlled Stall Demonstration

- Determine student level of comfortability with maneuvers thus far.
 - If the student is not feeling well, it is not advised to demonstrate a cross-controlled stall.
- Tell the student that this stall is for demonstration purposes and that they should not attempt to replicate it.

Power-On Stall

- First power-on stall will be a demonstration.
 - Instructor does, student says.
 - Verify correct procedure.
 - Points of Emphasis.
 - Smoothness.
 - Coordination.
- Second power-on stall will be performed by the student.
 - Student says, student does.
 - Verify correct procedure.
 - Provide feedback as needed throughout the maneuver
 - Points of Emphasis.
 - Smoothness.
 - Coordination.

Stall in Maneuver

- To demonstrate this, have the student put the airplane in slow flight.
- Once level, instruct the student to smoothly increase back pressure to induce a stall.
- Recover from the stall.

Recovery From Spiral Dive

- Instructor should put the airplane into a spiraling situation and instruct the student to recover.
 - Recovery:
 - Power idle.
 - Wings level.
 - Pitch up.

Turns Around a Point

- First completion of the maneuver will be a demonstration.
 - Instructor does, student says.
 - Verify correct procedure.
 - Points of Emphasis.

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- Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
- Second completion of the maneuver will be performed by the student
 - Student says, student does.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
 - Provide feedback as needed throughout the maneuver.

Normal Traffic Pattern

- The student should be able to establish and execute a proper traffic pattern with minimal instructor guidance.
- It will likely still be necessary for the instructor to provide guidance during the round out and flare portions.

Normal/Crosswind Landing

- The student should conduct the landing.
- Provide guidance during the round and flare.
- Provide guidance on proper crosswind correction.
 - Emphasize that this must be maintained even after touchdown.

Go-around/Rejected Landing

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 113.
- Assign homework:

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- Review the Flight Operations Manual (Version 6):
 - Part 02, Section 02 (Weather minimums for student pilot).
 - Part 03, Section 03 (Student pilot solo).
- Review the Pilot's Handbook of Aeronautical Knowledge:
 - Chapter 15 - Airspace.

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Conduct various stalls with instructor guidance.
4. Conduct traffic patterns, takeoffs and landings with instructor guidance.